

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

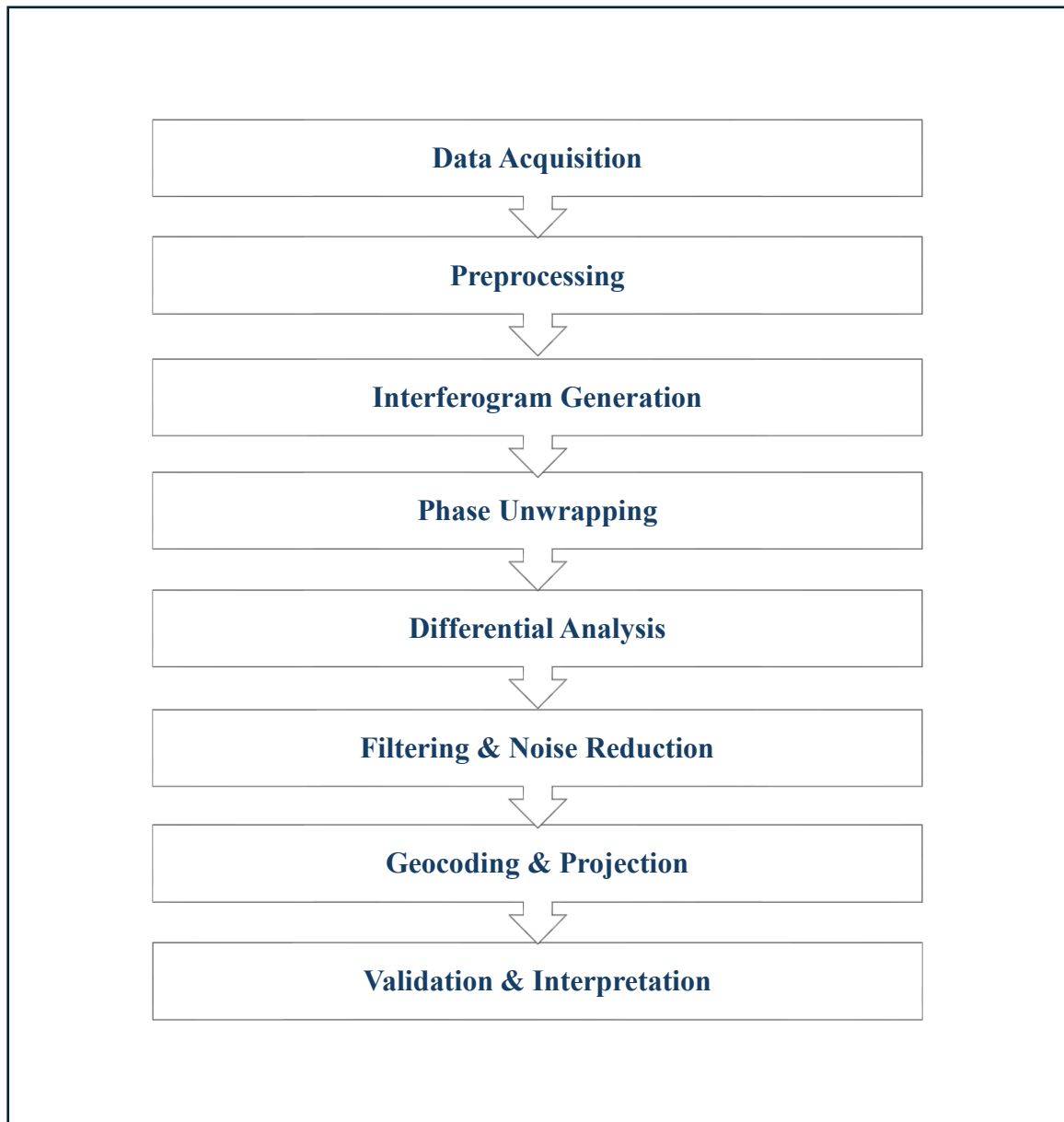
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

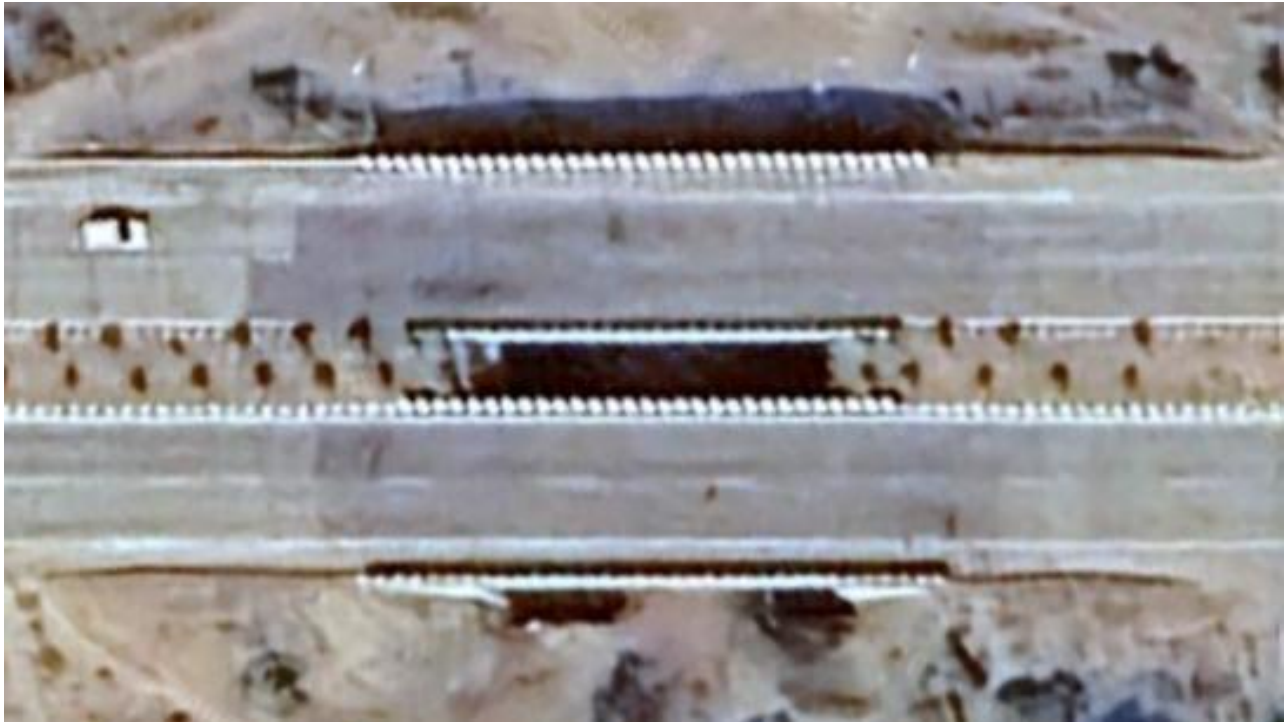
3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 899/1

Bridge Chainage – 965+680

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

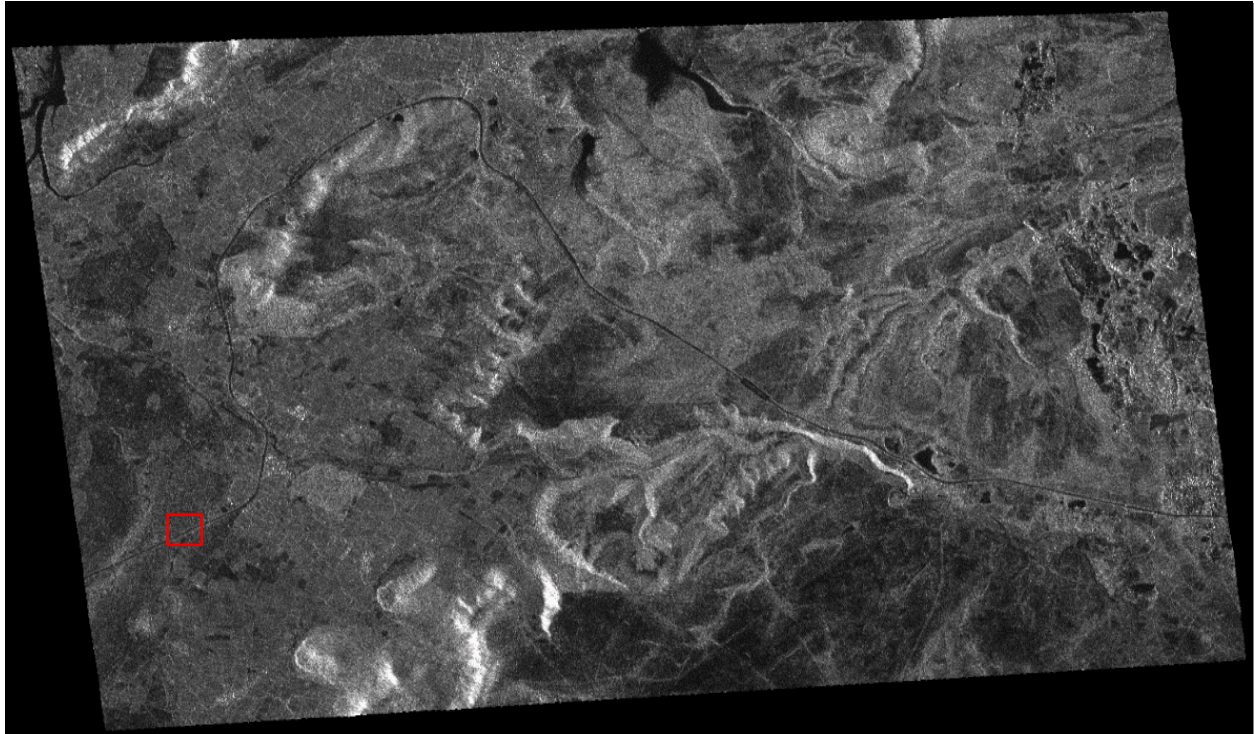
Flight Direction - DESCENDING

Polarization – VV+VH

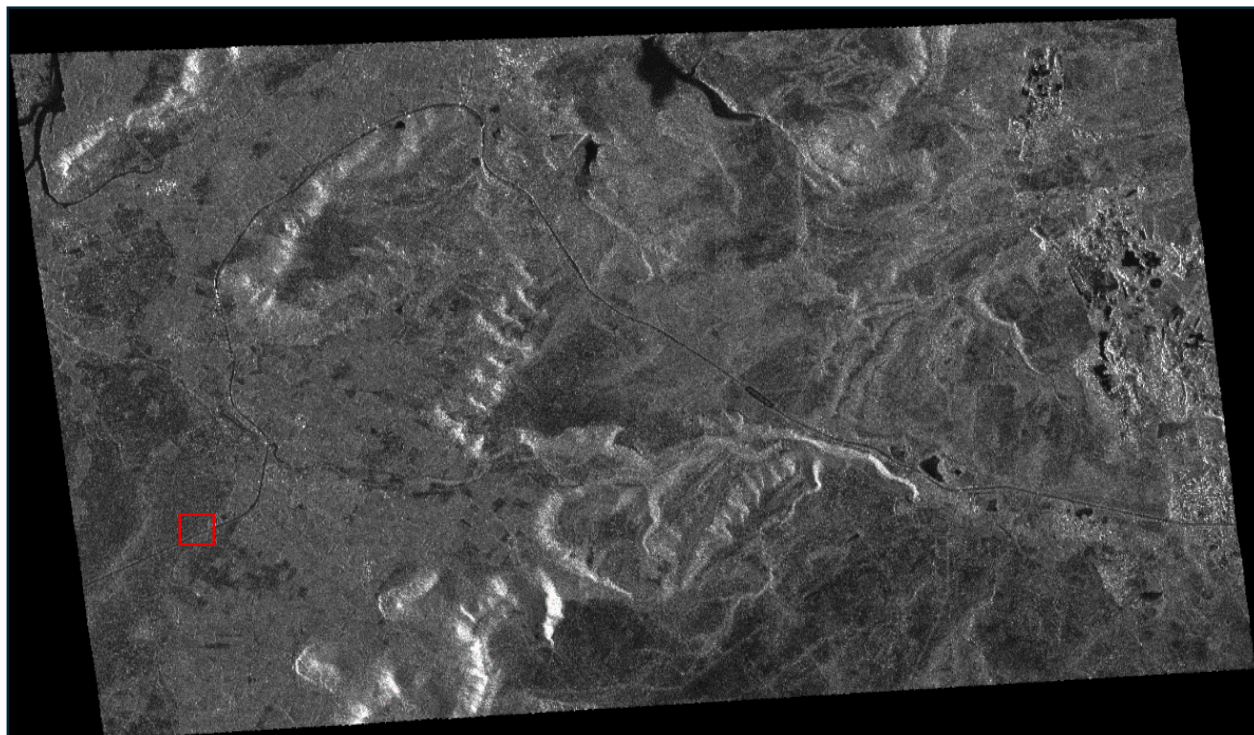
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

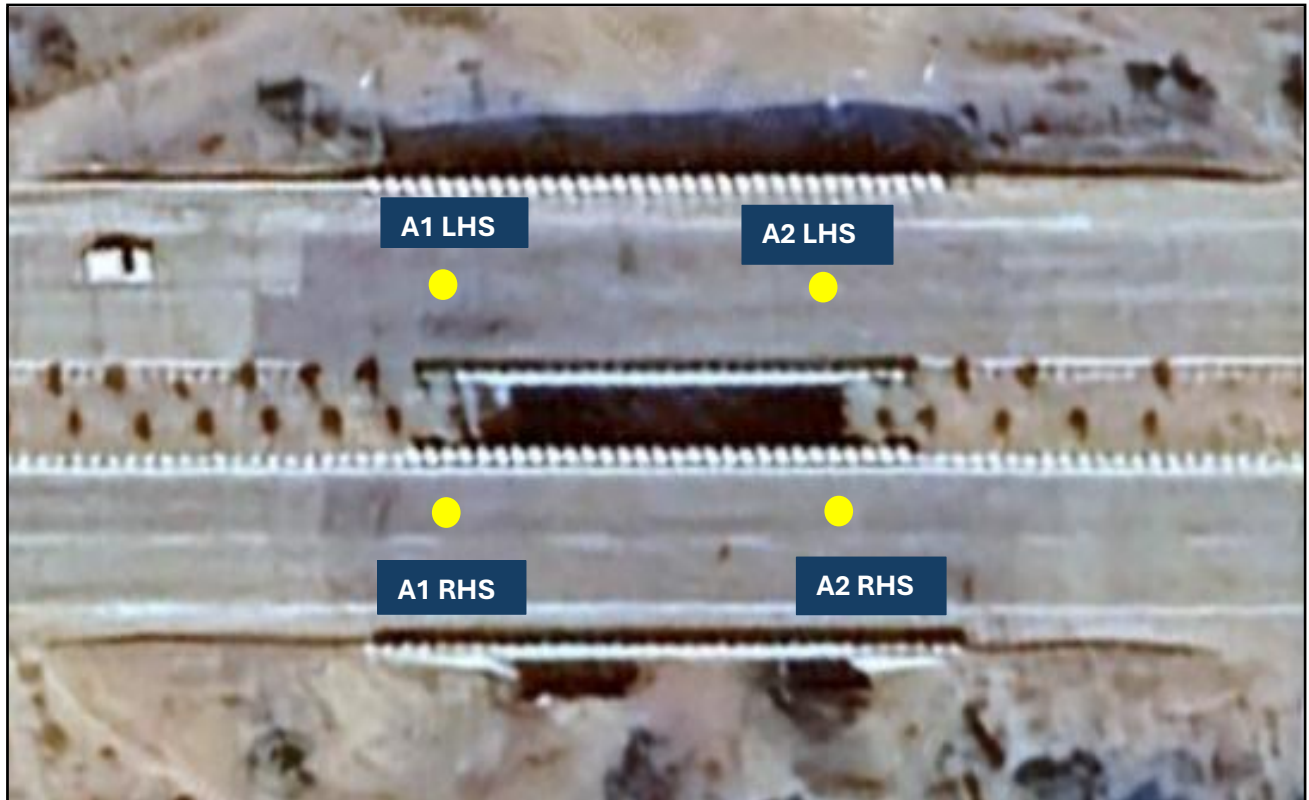


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

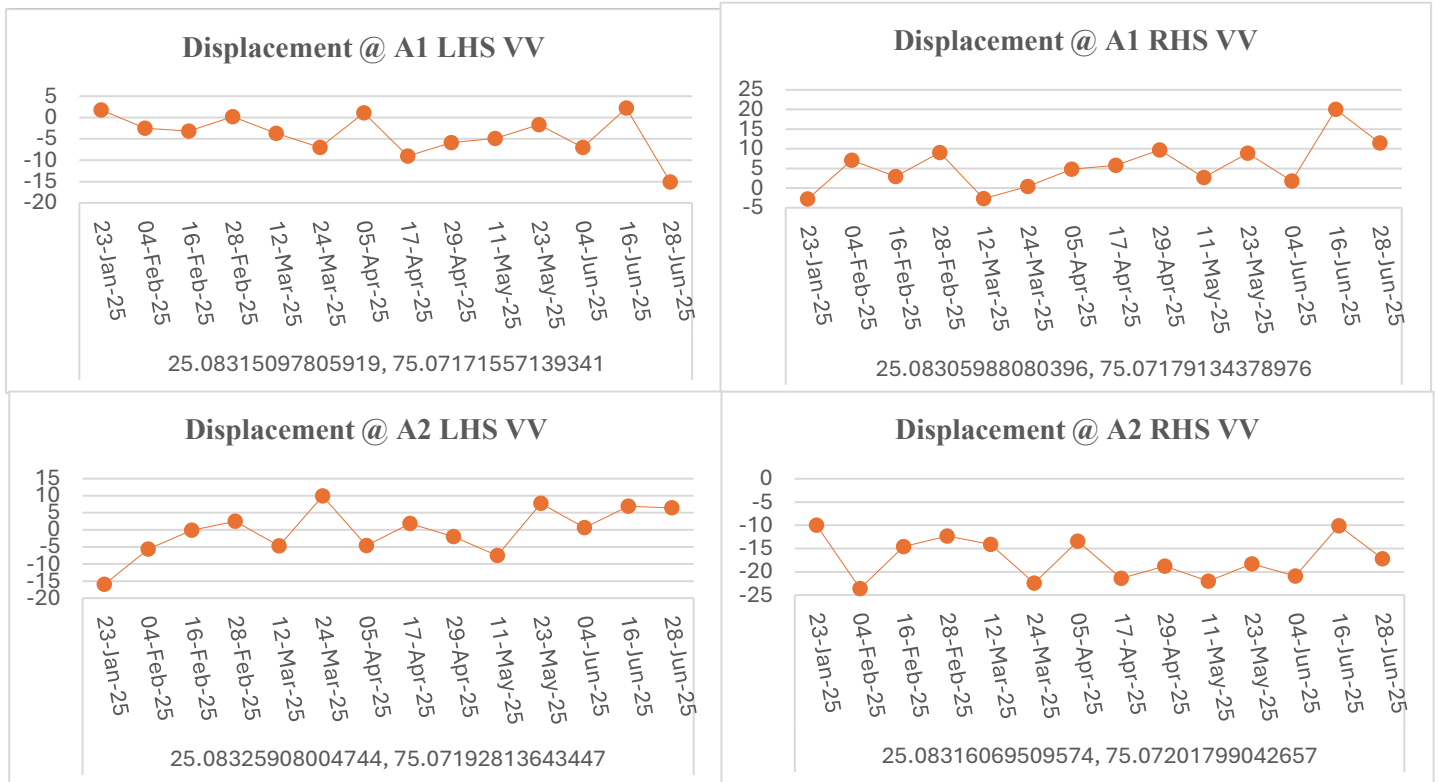
MNB – CH-965+680

Length – 34.69M

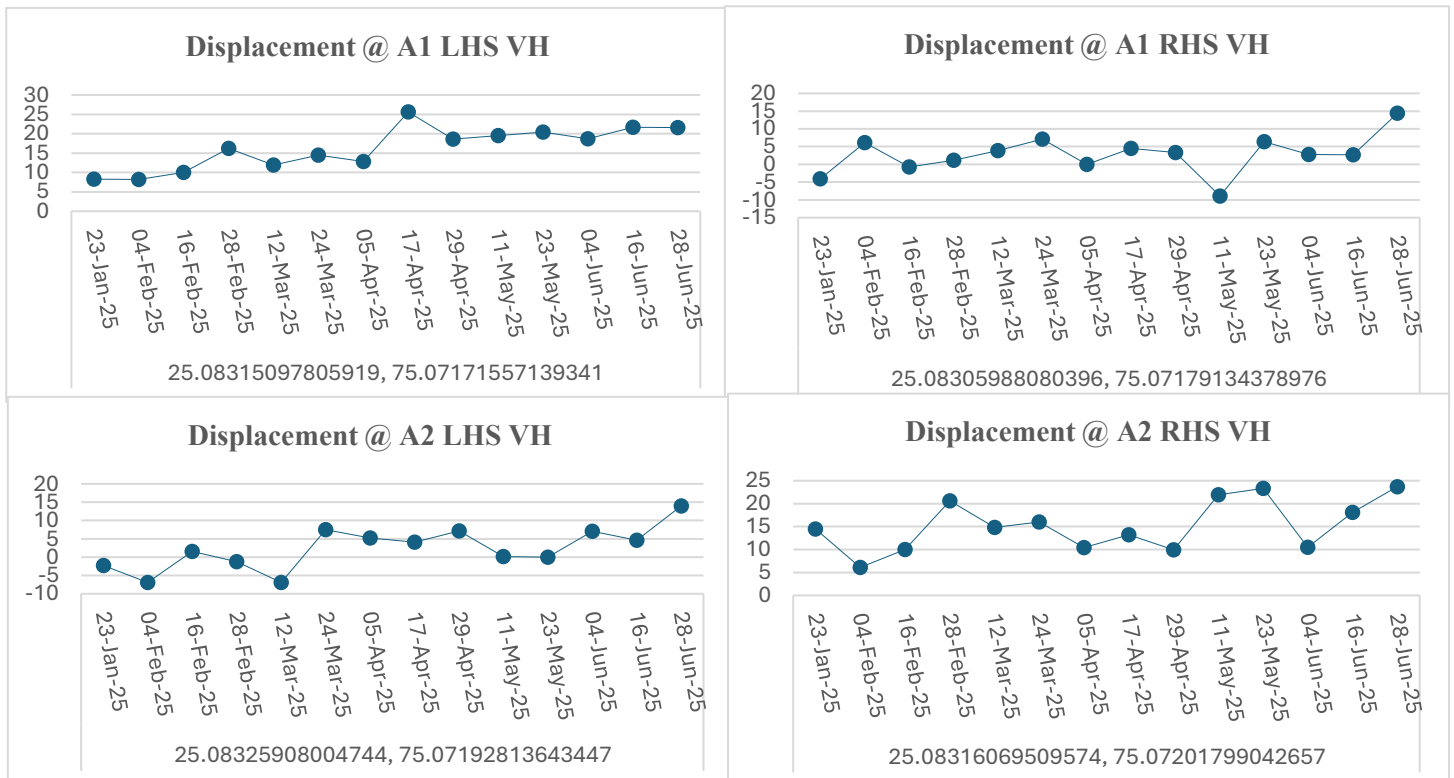


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1	Abutment 1		
A2	Abutment 2		
LHS	Left Hand Side		
RHS	Right Hand Side		

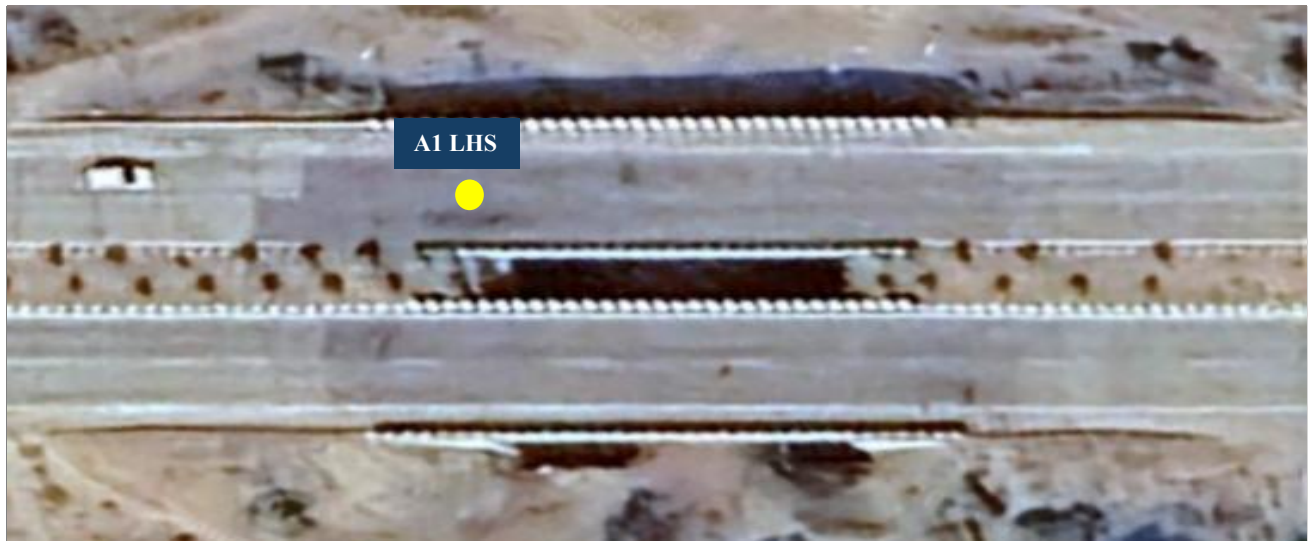
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



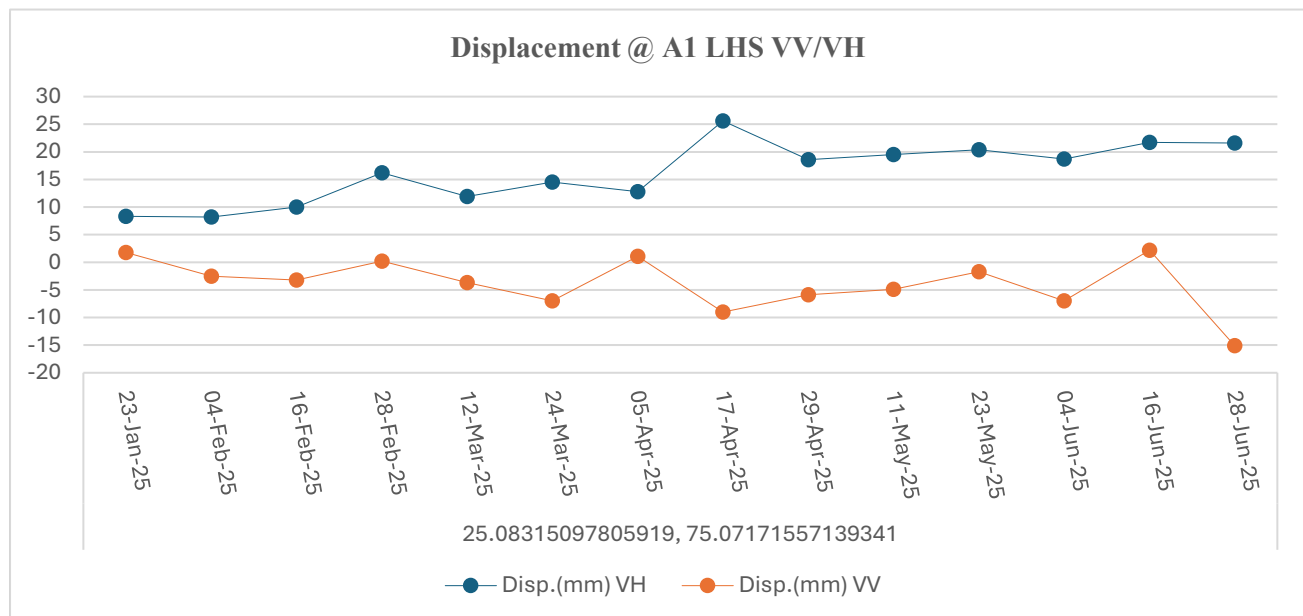
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (25.083162201418524, 75.07171080491364)



Key Observations:

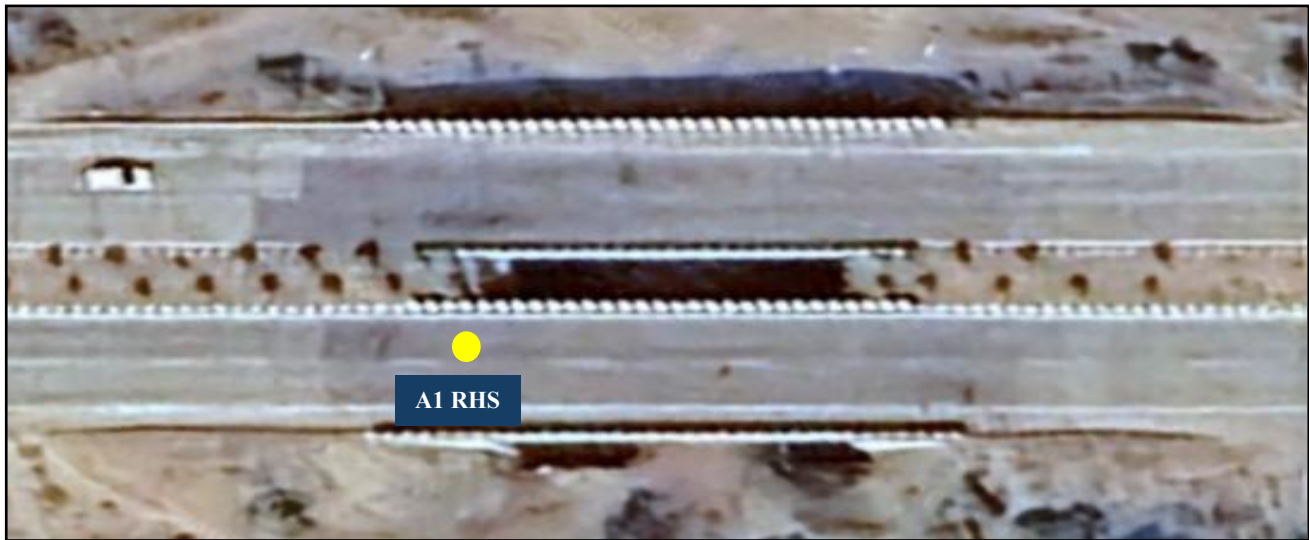
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** The vertical displacement exhibits alternating uplift and settlement behaviour throughout the monitoring period. Short-duration upward recoveries are observed between settlement phases, indicating elastic response rather than permanent deformation.
- **Horizontal displacement (VH)** Horizontal displacement shows a clear positive (northward) bias over time, increasing from roughly 8–10 mm in the early readings to about 18–22 mm in the later period. The trend is progressive and relatively smooth, with minor short-term oscillations, suggesting gradual lateral movement rather than abrupt shifts

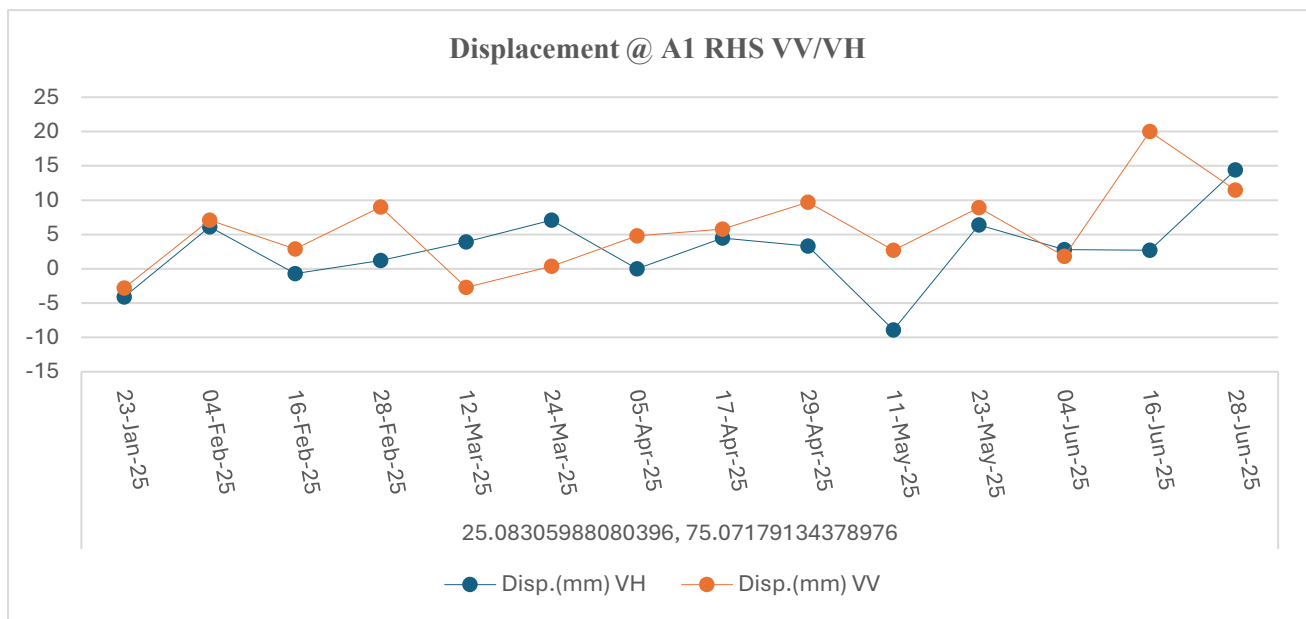
Critical Events & Anomalies:

- The pronounced vertical drop near the end of the monitoring window represents the most significant deviation and may be associated with transient loading effects, temperature variation, or local support response.
- No corresponding sudden spike in horizontal displacement is noted during this event, indicating that the movement is likely localized and not indicative of global instability.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (25.083048883684786, 75.07175704382105)



Key Observations:

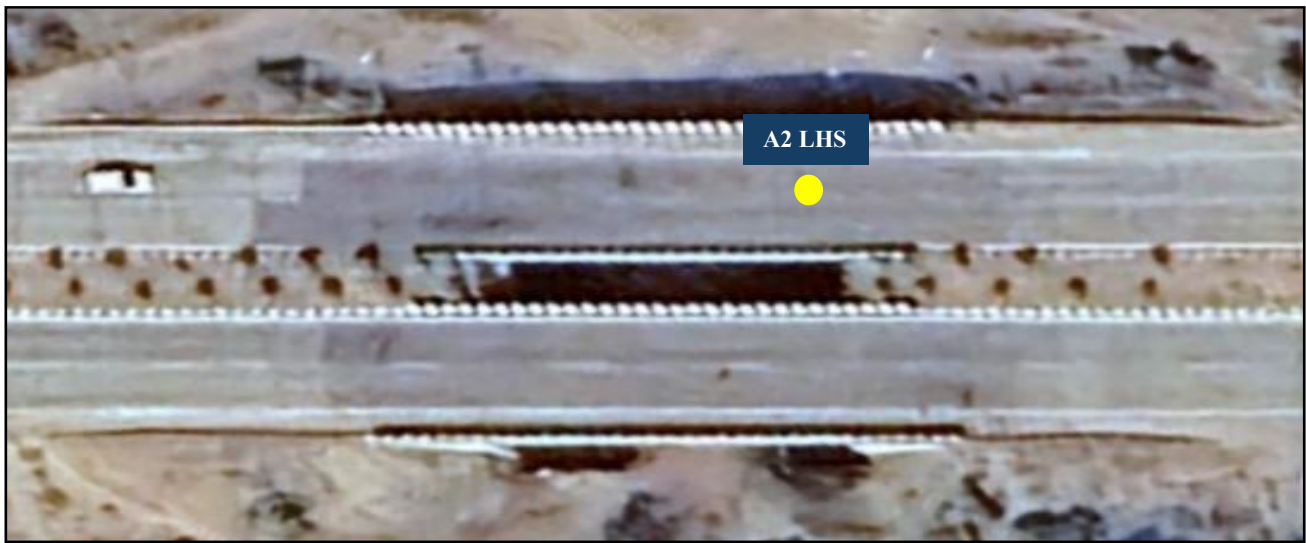
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement exhibits relatively small fluctuations around the neutral position, mostly confined between about -10 mm and +15 mm.
- **Vertical displacement (VV)** The vertical displacement response shows alternating uplift and downward movements, generally remaining within a moderate range (approximately -5 mm to +20 mm).

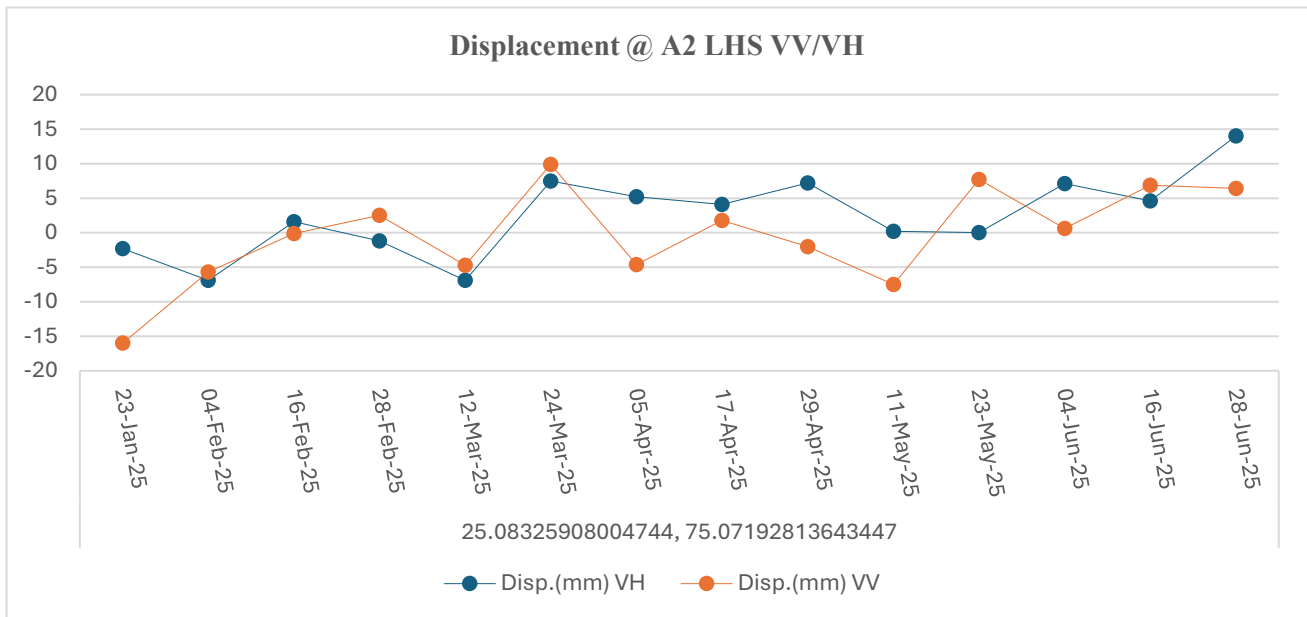
Critical Events & Anomalies:

- The isolated spike in vertical uplift and the short-lived negative horizontal movement appear transient and do not persist in subsequent readings.
- No cumulative increase or monotonic trend is observed in either vertical or horizontal displacement, indicating the absence of progressive settlement or lateral instability

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (25.083254169357193, 75.07190210706)



Key Observations:

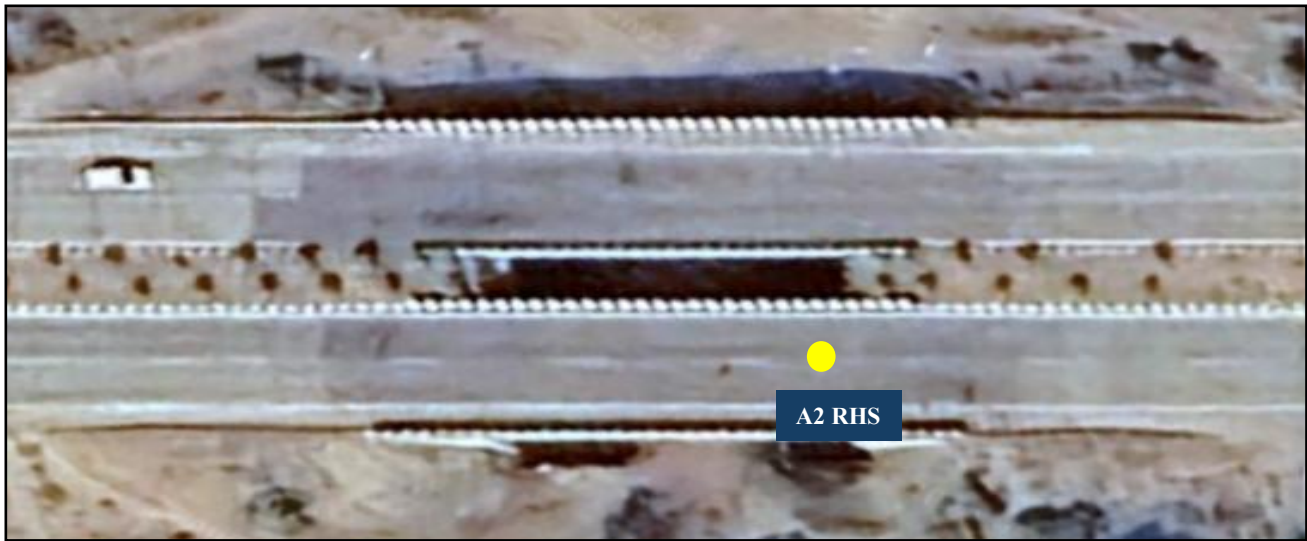
Displacement Magnitude Patterns:

- **Vertical displacement (VV)** Vertical displacements exhibit alternating upward and downward movements, generally ranging between approximately -15 mm and +10 mm.
- **Horizontal Displacement (VH)** Horizontal displacements remain mostly within -8 mm to +15 mm, showing moderate fluctuations over time. Initial movements indicate minor southward shifts, followed by a gradual transition towards northward displacement during later observations.

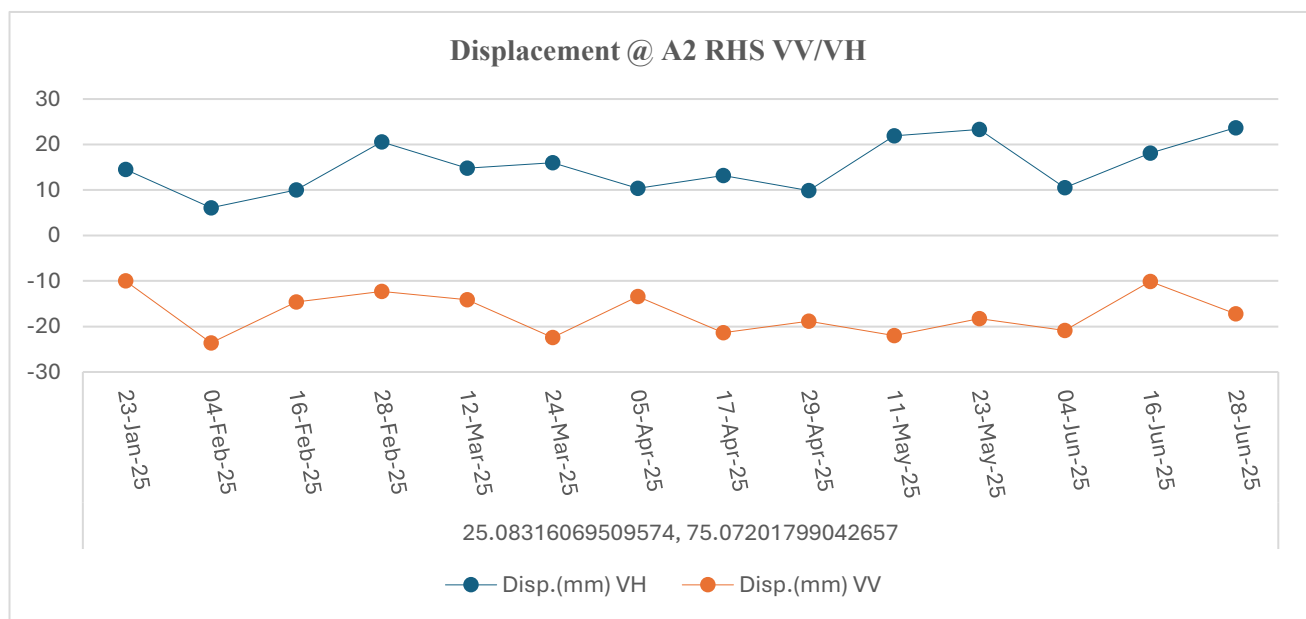
Critical Events & Anomalies:

- Short-term peaks in both VV and VH appear episodic rather than progressive, likely associated with transient loading or thermal effects.
- No abrupt or irreversible displacement jumps are observed, indicating the absence of sudden structural distress or foundation instability.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (25.083154811134744, 75.07196919880799)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacements generally remain within a moderate range, exhibiting both upward and downward movements over the monitoring period.
- **Horizontal (VH) displacement** Horizontal displacements show more pronounced variability compared to vertical movements, with values typically ranging from about 0 mm to +30 mm in several monitoring locations.

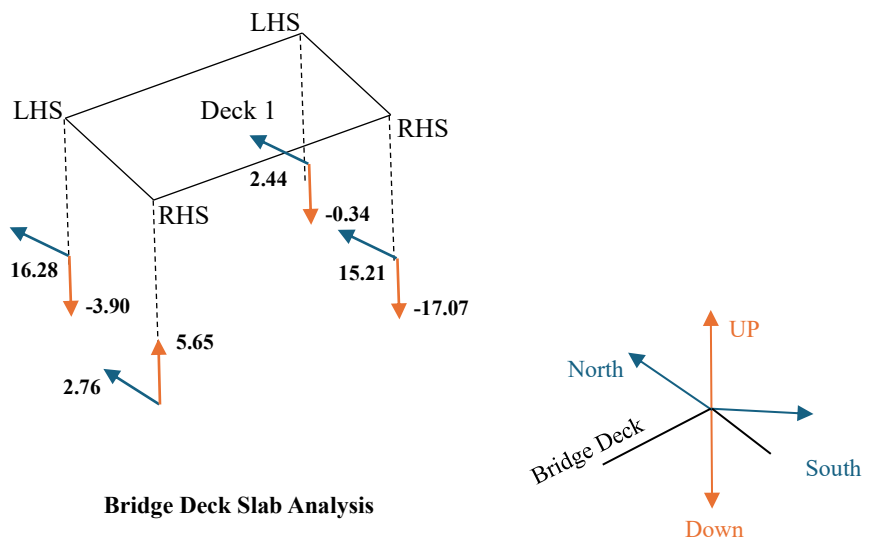
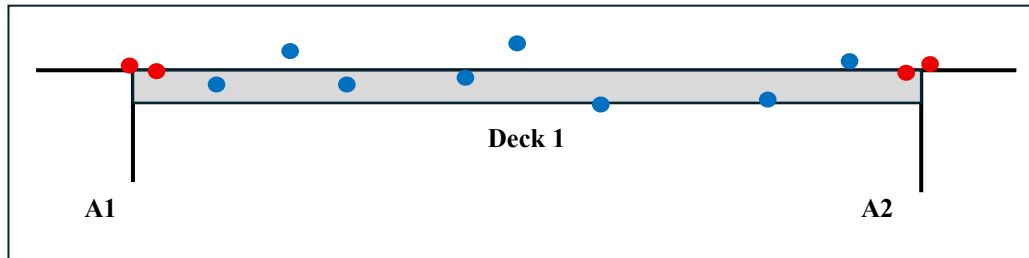
Critical Events & Anomalies:

- Isolated short-term drops in vertical displacement (negative VV values) appear at specific epochs; however, these are followed by recovery in subsequent readings, indicating no progressive settlement trend.
- Horizontal displacement peaks occur intermittently but remain consistent with expected seasonal and operational effects rather than structural distress. The absence of simultaneous extreme vertical and horizontal displacement at the same epoch further supports stable structural behaviour.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- **Points of Analysis (POA) of bridge (●)**



Vertical Displacement Behaviour:

- The vertical displacements observed across Deck 1 show a combination of minor uplift and downward movement, with values remaining within acceptable serviceability limits for bridge deck slabs. Downward movements are moderate and localized, while the recorded upward displacements indicate normal elastic recovery of the slab under varying thermal conditions and traffic loading.

Horizontal Displacement Behaviour:

- Horizontal displacements exhibit small to moderate magnitudes, reflecting typical thermal expansion-contraction behaviour of the bridge superstructure. The lateral movements vary in direction between LHS and RHS, but remain well controlled and non-cumulative, indicating effective lateral restraint and absence of deck-level drift.

CONCLUSION:

- The combined vertical and horizontal displacement patterns for Deck 1 indicate normal and stable structural behaviour. All measured movements fall within acceptable engineering limits and show no signs of abnormal deformation, distress, or progressive instability. The bridge deck slab is performing satisfactorily under current service conditions, and continued routine monitoring is sufficient to track long-term performance trends